

Calibration procedure using a gashood

Guide Application Note

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Revision history

Date	Revision	Changes
08/11/2024	0.0	Initial Release

Table 1 – Change log

1. GASHOOD

To perform an accurate calibration of the gas cartridges or a bump test, a gashood (Figure 1 – Gashood for AIR Pro (K-GASHOOD-AP-01) and for AIR Lite (K-GASHOOD-AL-01)) can be used to connect a certified calibration gas cylinder to the Kunak AIR station. This type of calibration is similar to the calibration protocol for reference instruments.



Figure 1 – Gashood for AIR Pro (K-GASHOOD-AP-01) and for AIR Lite (K-GASHOOD-AL-01)

2. INITIAL CONSIDERATIONS FOR CALIBRATION

Before starting the calibration process, ensure the following:

- 1) Power on the Kunak AIR station and allow the gas sensors to stabilize (>5 days).
- 2) Set the sensor sampling period to measure at least once every 30 seconds. To adjust the sampling period remotely:
 - On the KunakCloud website, navigate to: Configuration > Simple > Sensor
 - Select the device and sensor to modify.
 - Set the desired interval in Sampling > Period.
 - Click Save to apply the changes.
 - The new sampling period will take effect after the device's next communication with the server.

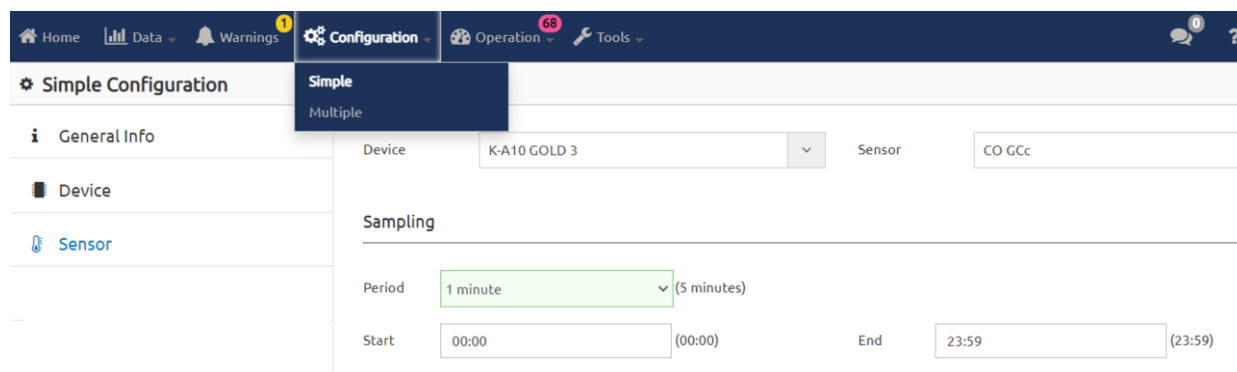


Figure 2 – Adjusting the sampling period in KunakCloud

- 3) Adjust the device sending period to avoid electromagnetic interference. Cellular communications can temporarily affect sensor readings, as the equipment pauses

measurement for 1 to 5 minutes during the transmission of the data to KunakCloud. For calibration, set the sending period to at least 15 minutes to minimize interruptions. Consider exposing the sensors to the calibration gas 2-5 minutes after the communication period of the device with Kunak Cloud, to avoid the loss of data during the **guard period***.

To ensure maximum data reliability, the device includes a data discrimination feature to mitigate electromagnetic interference (EMI) that may affect measurements during transmission to the Kunak Cloud. After each connection to the Kunak Cloud, the device **discards data collected during the following 2 to 5 minutes. This interval, known as the **guard period**, varies depending on the configured sampling and sending periods.*

To change the sending period remotely:

- On the KunakCloud website, go to: Configuration > Simple > Device
- Select the device to modify.
- Update the interval in Communications > Sending period.
- Click Save to send the change to the device.
- The new sending period will apply after the next data transmission.

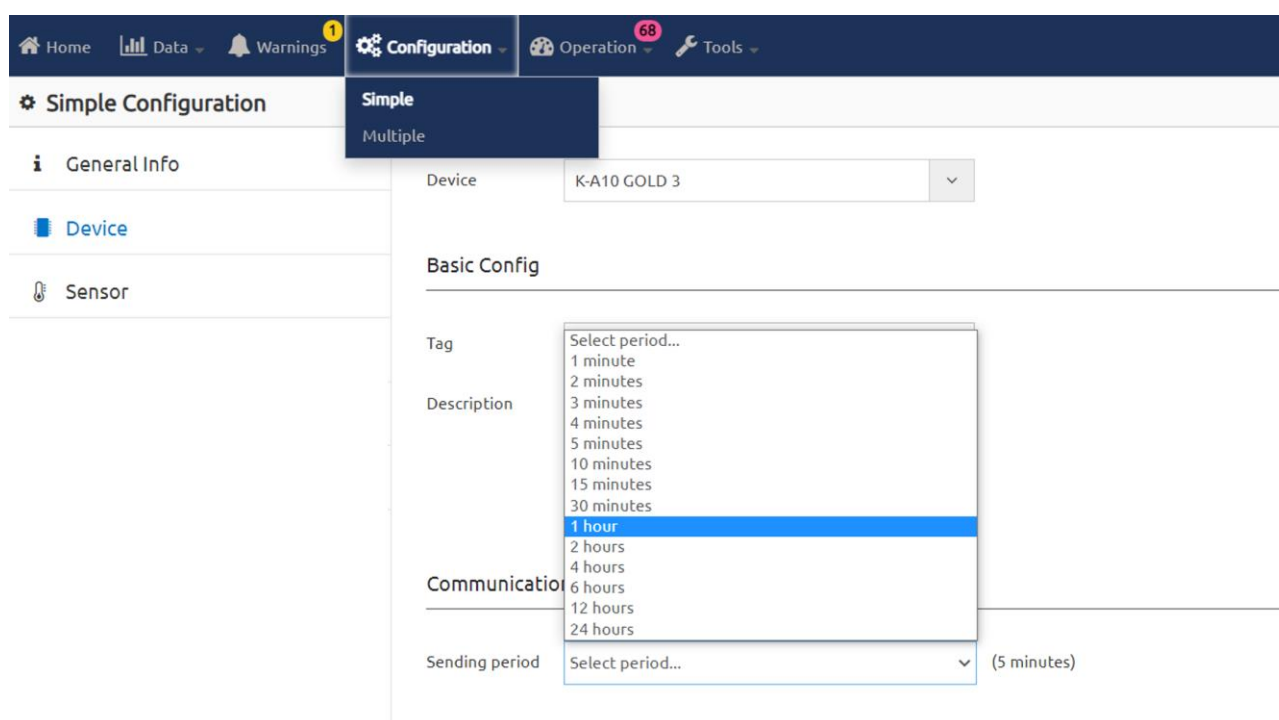


Figure 3 – Adjusting the sending period

- 4) The temperature, humidity, and pressure during calibration should match the expected conditions at the device's final location. If these conditions are unknown, use the recommended values listed in Table 2.

Parameter	Operating Range	Recommended Testing Values
Temperature	-30 °C to 40 °C	20 °C
Humidity	15 %RH to 85 %RH	50 %RH
Pressure	80 KPa to 120 KPa	101.325 KPa

Table 2 – Temperature, humidity and pressure conditions for calibration

It is recommended to avoid **sudden humidity changes whenever possible**. In situations where abrupt RH shifts are unavoidable, such as when connecting a gas cylinder with a different RH from ambient, the **resulting spikes in sensor readings should be disregarded**.

Avoid using dry air (refer to Table 1) during calibration, except for VOC sensor calibration, as it can affect the accuracy of electrochemical sensors. Ensure that humidity levels remain stable throughout the entire calibration process to maintain consistent results.

To guarantee optimal performance of the Kunak Air system, two types of calibration can be performed: **Baseline Calibration** and **Span Calibration**. Follow the steps below carefully.

3. BASELINE CALIBRATION (ZERO CALIBRATION)

This step sets the zero value for the sensors. It must always be performed **before Span Calibration**. When calibrating electrochemical sensors, use **zero air** at 50% relative humidity for the best results. Note that zero air with a different RH may cause small baseline variations in electrochemical sensors (on the order of a few ppb). For the VOCs cartridges based on PID sensor, dry gas would be preferred.

Procedure:

1. Place the gashood over the sensors and secure it by tightening the black screws on the hex standoffs of the Kunak station by hand.



Figure 4 - Installation gashood on Kunak AIR Pro

2. Connect the zero air source (gas cylinder or generator) to the gashood inlet port using a 6 mm PTFE tube and the inlet quick connector of the port. Remember that the length of the PTFE tube will affect the time it takes for the calibration gas to reach the sensors.



Figure 5 - Connection of gas cylinder to the gashood

3. Introduce zero air at a **flow rate of 1–1.5 L/min for at least 15 minutes**. The total duration will depend on the length of the tubing, as the ambient air inside the tube must be fully purged before zero air reaches the sensors.
4. Stop the gas flow and disconnect the PTFE tube from the gashood inlet port.
5. Remove the gashood by untightening the black screws by hand.
6. Use the **KunakCloud Calibration Tool** to calibrate the sensitivity of the sensors following the instructions in the [user manual](#).



NOTE: The sensors accuracy at zero concentration is expected to be around ± 5 ppb for NO, NO₂, O₃, H₂S and SO₂ and ± 20 ppb for CO if the steps have been correctly followed and no humidity transients are present during the calibration procedure.

All sensors should have their baseline calibrated before performing the span calibration.

4. SPAN CALIBRATION (SENSITIVITY CALIBRATION)

This procedure ensures the correct sensitivity of the sensors by calibrating their response to a known concentration of gas.

Procedure:

1. Place the gashood over the sensors and secure it by tightening the black screws on the hex standoffs of the Kunak station by hand.

2. Connect the gas cylinder to the gashood inlet port using a 6 mm PTFE tube and the inlet quick connector of the port.

**RECOMMENDATIONS ON GAS CYLINDERS:**

- Different types of gas cylinders may exhibit varying error tolerances, which should be considered by the user based on the intended application.
- Lecture bottles and disposable cylinders: typically have an error tolerance of 10%. Recommended for verification.
- Large gas cylinders: generally, possess a lower error tolerance (1- 2%). Recommended for calibration.

3. Introduce a constant gas concentration at a **flow rate of 1–1.5 L/min for at least 15 minutes**. Ensure the tubing is fully purged to eliminate any residual ambient air.
4. Recommended gas concentrations:
 - **500 ppb – 1 ppm:** NO, NO₂, O₃, H₂S, SO₂, H₂S Type A
 - **5–10 ppm:** CO Type A
 - **10–400 ppm:** CO Type B
 - **2 ppm:** VOCs Type A (Use isobutylene for VOC calibration)
 - **10–20 ppm:** VOCs Type B (Use isobutylene for VOC calibration)
 - **5–40 ppm NH₃ Type A and C**
 - **10 – 1200 ppm NH₃ Type B**
 - **5–15 ppm: H₂S Type B**
5. Stop the gas flow and disconnect the PTFE tube from the gashood inlet port.
6. Remove the gashood by loosening the black screws by hand.
7. Use the **KunakCloud Calibration Tool** to calibrate the sensitivity of the sensors following the instructions in the [user manual](#).

**ADDITIONAL NOTES:**

- The baseline calibration must be done before the span calibration.
- Ensure all tubing connections are airtight to avoid leaks and ensure accurate calibration.
- Avoid contaminants in the enclosure, especially when calibrating O₃ and NO_x, as these can impact sensor accuracy.